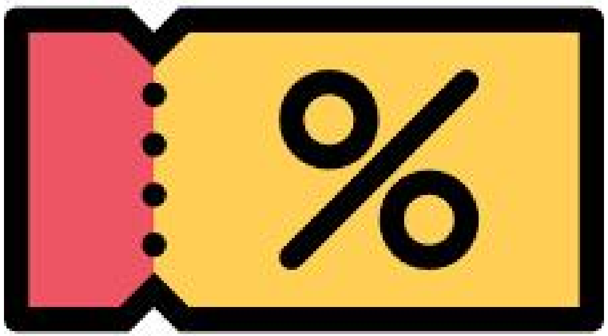
Predictive Modeling for Coupon Acceptance: A Business Application of DNN



Presented by Team 26:

XU Yibin
ZHAN Shuqing

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Objective

Business Problem

- Companies frequently use coupons as a marketing tool with the aims to attract new customers, retain the existing ones, and stimulate increased spending.
- The distribution of coupons, however, could be expensive and its effectiveness dependent largely on the acceptance and utilization by potential customers.

The objectives of DNN application are to

✓ predict **whether a person will accept a coupon** under various scenarios and personal attributes.
✓ strategically, guide the business's coupon distribution, ultimately optimizing marketing resources and increasing overall return on investment (ROI).
✓ tactically, help the business target customers who are more likely to accept the coupons, leading to enhanced customer engagement and sales.

Dataset

Source and Volume

- Dataset source: UCI Machine Learning Repository
- Collected via Amazon Mechanical Turk survey in 2017
- Comprises 12k instances with a balanced outcome (57% acceptance, 43% non-acceptance).

Features Overview

25 categorical features divided into four categories:

- **Scenario Factors**: destination, current time, weather, type of coupon, coupon expiration time, and car passengers.
- **Personal Attributes**: gender, age, marital status, presence of children, education, occupation, and income.
- **Activity Frequency**: frequency of visits to a bar, coffeehouse, and two types of restaurants (below $20 per person and $20 to $50 per person).
- **Coupon-Related Factors**: driving distance to coupon-use location (more than 15 or 25 minutes) and if location is in current travel direction.

Data Split for Modeling

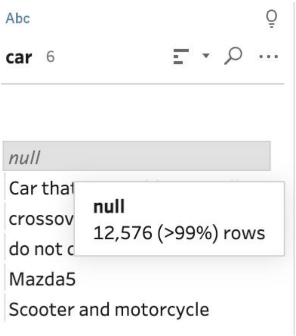
•Training dataset: 70% (8.4k instances)
•Validation dataset: 20% (2.4k instances)
•Test dataset: 10% (1.2k instances)

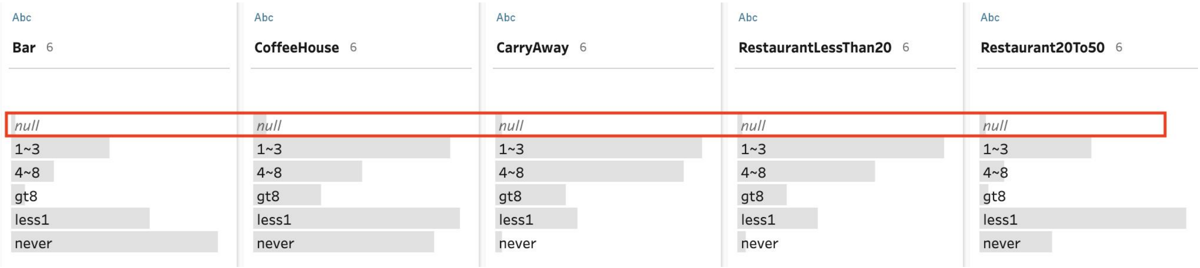
Data Processing(1/4)

Number of features: 25 -> 24

Missing Value Handling → Elimination of Feature with Single Value → Categorical Variables Conversion → Elimination of Perfectly Correlated Features

1. Eliminated the “car” feature due to almost exclusive NA values.
2. Removed 794 data entries with missing values in five specific features, keeping the features intact.



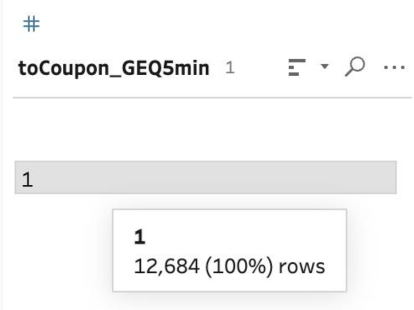


Data Processing(2/4)

Number of features: 24 -> 23

Missing Value Handling → **Elimination of Feature with Single Value** → Categorical Variables Conversion → Elimination of Perfectly Correlated Features

Dropped “toCoupon_GEQ5min” as it contain a single value “1”.



Data Processing(3/4)

Number of features: 23 -> 63

Missing Value Handling → Elimination of Feature with Single Value → **Categorical Variables Conversion** → Elimination of Perfectly Correlated Features

1. Converted 8 categorical features into **numerical variables** based on their intuitive meaning (e.g., "never" to "0").

Variables below were converted from categorical to numerical variables:

```
'temperature', 'age','income', 'Bar', 'CoffeeHouse', 'CarryAway',
'RestaurantLessThan20', 'Restaurant20To50'
```

Mapping between categories and corresponding numbers.

```
temperature_value_map = {'30':30, '55':55, '80':80}
age_value_map = {'below21': 21, '21':21, '26':26, '31':31, '36':36, '41':41, '46':46, '50plus': 50}
income_value_map = {'Less than $12500':12500,
                    '$12500 - $24999':18750,
                    '$25000 - $37499':31250,
                    '$37500 - $49999':43750,
                    '$50000 - $62499':56250,
                    '$62500 - $74999':68750,
                    '$75000 - $87499':81250,
                    '$87500 - $99999':93750,
                    '$100000 or More':100000}
Bar_value_map = {'never':0,'less1':1,'1~3':2, '4~8':6, 'gt8':8}
CoffeeHouse_value_map = {'never':0,'less1':1,'1~3':2, '4~8':6, 'gt8':8}
CarryAway_value_map = {'never':0,'less1':1,'1~3':2, '4~8':6, 'gt8':8}
RestaurantLessThan20_value_map = {'never':0,'less1':1,'1~3':2, '4~8':6, 'gt8':8}
Restaurant20To50_value_map = {'never':0,'less1':1,'1~3':2, '4~8':6, 'gt8':8}
```

2. Transformed remaining 15 categorical variables into **dummy variables**.

Variables below were converted from categorical to dummy variables:

```
'destination', 'passanger', 'weather', 'time', 'coupon', 'expiration', 'gender',
'maritalStatus', 'has_children', 'education', 'occupation', 'toCoupon_GEQ15min',
'toCoupon_GEQ25min', 'direction_same','direction_opp'
```

Data Processing(4/4)

Number of features: 63 -> 61

Missing Value Handling → Elimination of Feature with Single Value → Categorical Variables Conversion → **Elimination of Perfectly Correlated Features**

We examined the **correlation matrix** of our dataset and found that

- the feature “time_7AM” was perfectly correlated (correlation of 1) with “destination_Work”
- the feature “direction_opp_1” was perfectly inversely correlated (correlation of -1) with “direction_same_1”.

These correlations indicate redundancy as one feature can be linearly predicted from the other. Therefore, we **removed one feature from each of these two pairs**, specifically "destination_Work" and "direction_opp_1", to eliminate this redundancy.

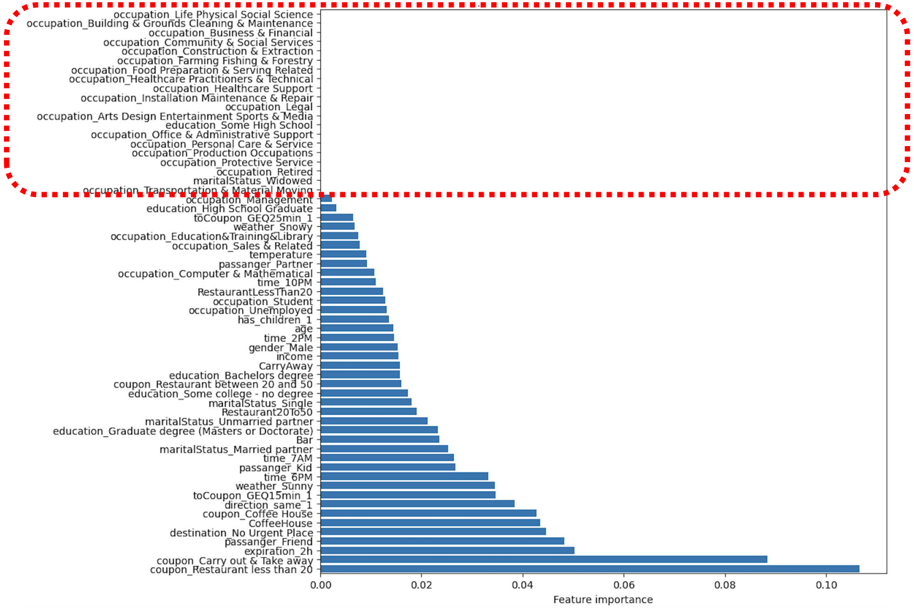
By the end of the data processing stage, we had a refined dataset with a total of **61** features. We then did **data normalization** for numerical features before the subsequent features selection step.

Feature Selection (1/2)

Feature Importance Measure from XGBoost → Test Different Number of Features

Eliminate due to 0 importance

Feature Importances in XGBoost Classification Model



- The feature selection process starts with the tuning of an XGBoost Classifier on the training set to identify and evaluate **feature importance**.
- Feature importance is determined based on the impact of each feature on the **model's predictive accuracy**.
- In this step, we discovered that out of the total features, **20 had zero importance**, meaning they contributed nothing towards the model's predictive ability.
- These non-essential features were dropped, leading to a refined set of **41 significant features** that were kept for further utilization.

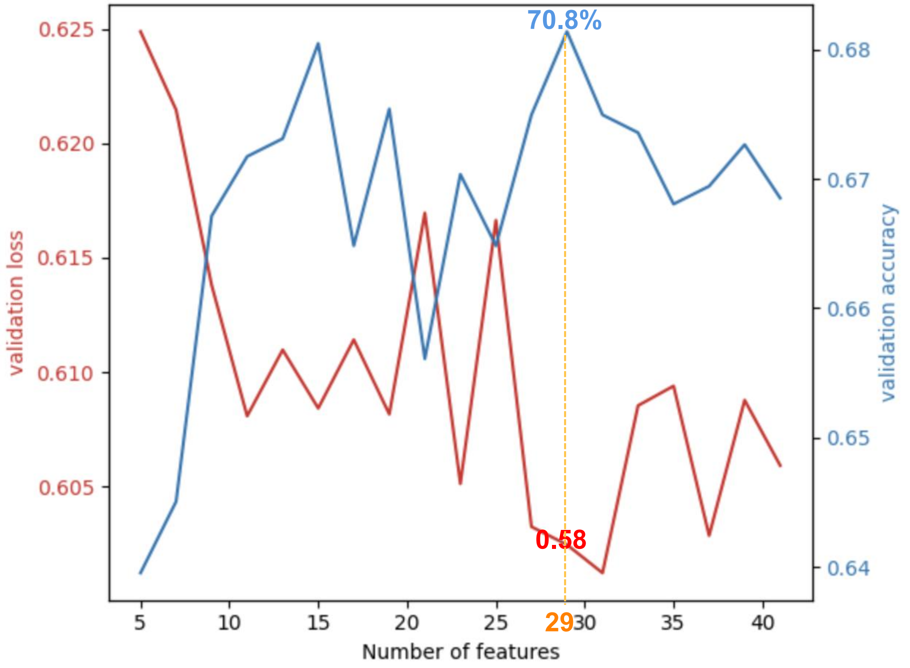
Feature Selection (2/2)

Feature Importance Measure from XGBoost → Test Different Number of Features

Baseline Model: DNN(B)-frs(5240)-[12(relu)-8(relu)-1(sigmoid)]-[ep(30)]

- Develop a **baseline model** and **experiment with different numbers of features** within this model.
- Feature examination was conducted based on their importance rankings. For example, a model with 20 features would incorporate the top 20 features as ranked by their importance.
- The optimal number of features - the count that yielded the **highest validation accuracy** - was selected for further modeling.
- As shown in the figure on the right, the optimal number of features is **29**, which brings the highest validation accuracy on the baseline model.

Accuracy and loss with different number of features

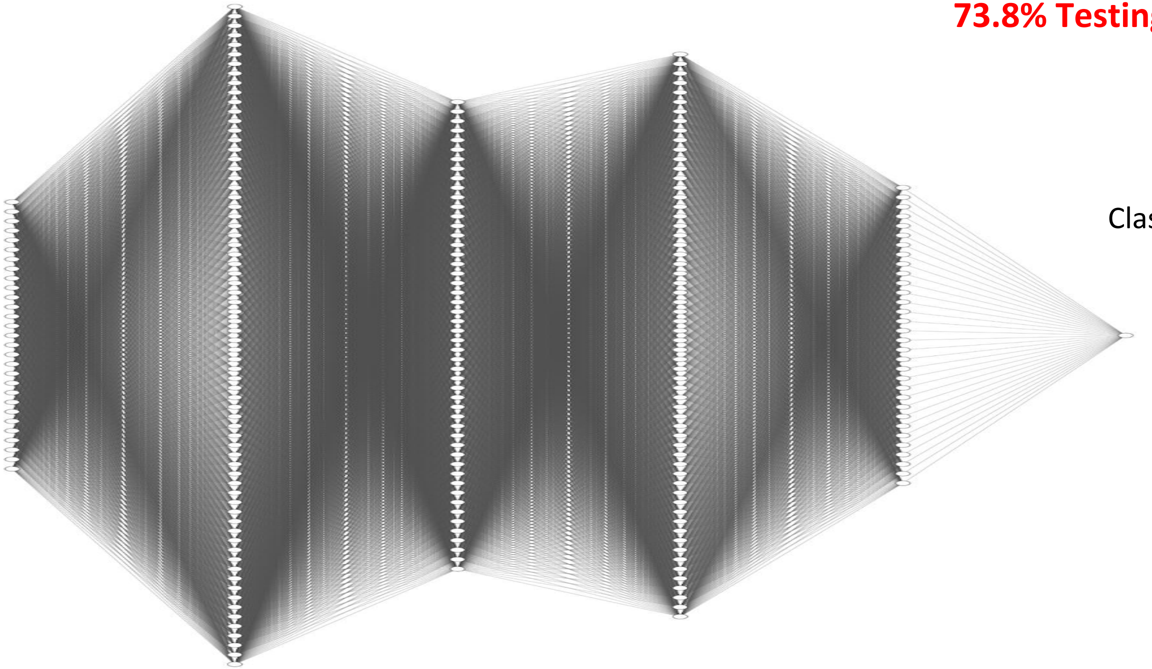


Model Structure

Best Model: DNN(B)-frs(5240)-[135(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]

73.8% Testing Accuracy

Input **29** features



135 Nodes
Activation: Relu

85 Nodes
Activation: Relu

94 Nodes
Activation: Relu

32 Nodes
Activation: Relu

Classification Outcome

Accept (1)
OR
Not Accept (0)

As there are too many neurons,
the graph is for illustration purpose.

Experiments

- **Baseline Model** Our baseline model is defined as DNN(B)-frs(5240)-[12(relu)-8(relu)-1(sigmoid)]-[ep(30)], and our experimentation begins from here.

- **Initial Experiment**:
- Broad-range adjustment of layers and neurons.
- Selection of the highest accuracy and lowest loss model as the baseline.

- **Experiment Details**

1. **Architectural Refinements**: Extensive adjustment of layers and neurons.
2. **Hidden Layer and Batch Size Adjustments**: Different neuron counts tested in the first and third hidden layers for batch sizes of 20 or 40.
3. **Epoch Optimization**: Further testing of top-performing models for different epochs.
4. **Overfitting Mitigation and Generalization Improvement**: Incorporation of L1 regularization and Dropout to compare with the baseline.
5. **Model Refinement and Testing**: Continuous monitoring of model performance and refinement of the optimal model.

Performance

Among our current experiments(71 in total), 3 DNN models that achieved the highest test accuracy are shown as below.

Index	layer	epochs	batch_size	base: Los	base: Accurac	Best resul	L1: Los	L1: Accurac	Best resul	dropout: Los	dropout: Accurac	Best resul
47	DNN(B)-frs(5240)-[105(relu)-85(relu)-109(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.142	73.18%	best result	0.614	70.78%		0.614	71.61%	
68	DNN(B)-frs(5240)-[135(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.300	70.94%		0.614	70.86%		0.608	73.76%	best result
3	DNN(B)-frs(5240)-[30(relu)-12(relu)-24(relu)-12(relu)-8(relu)-1(sigmoid)]-[ep(30)]	30	40	0.593	69.54%		0.612	72.02%	best result	0.689	57.70%	
39	DNN(B)-frs(5240)-[135(relu)-85(relu)-64(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.187	73.10%		0.618	70.12%		0.608	71.85%	
34	DNN(B)-frs(5240)-[120(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.194	72.43%		0.612	71.44%		0.611	72.76%	
14	DNN(B)-frs(5240)-[90(relu)-85(relu)-64(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.234	71.69%		0.609	70.28%		0.610	71.85%	
61	DNN(B)-frs(5240)-[135(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.036	71.52%		0.626	70.20%		0.611	71.69%	
22	DNN(B)-frs(5240)-[105(relu)-85(relu)-64(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.106	71.44%		0.615	71.36%		0.611	71.94%	
35	DNN(B)-frs(5240)-[120(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.094	71.36%		0.619	70.20%		0.612	72.10%	
16	DNN(B)-frs(5240)-[90(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.090	71.36%		0.600	71.52%		0.612	72.02%	
45	DNN(B)-frs(5240)-[135(relu)-85(relu)-109(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.300	71.27%		0.616	70.03%		0.612	71.85%	
30	DNN(B)-frs(5240)-[120(relu)-85(relu)-64(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.196	71.27%		0.611	70.36%		0.610	72.19%	
23	DNN(B)-frs(5240)-[105(relu)-85(relu)-64(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.150	71.27%		0.611	69.78%		0.609	71.94%	
25	DNN(B)-frs(5240)-[105(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.100	71.19%		0.617	69.45%		0.615	71.85%	
54	DNN(B)-frs(5240)-[120(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.077	71.19%		0.615	70.61%		0.609	72.02%	
17	DNN(B)-frs(5240)-[90(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.116	71.03%		0.613	70.36%		0.614	71.36%	
43	DNN(B)-frs(5240)-[135(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.232	70.94%		0.615	69.12%		0.612	73.01%	
63	DNN(B)-frs(5240)-[135(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.292	70.86%		0.619	69.70%		0.611	72.19%	
51	DNN(B)-frs(5240)-[120(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.105	70.86%		0.612	71.27%		0.618	71.11%	
2	DNN(B)-frs(5240)-[20(relu)-8(relu)-12(relu)-1(sigmoid)]-[ep(30)]	30	40	0.565	70.86%		0.600	71.27%		0.689	57.70%	
65	DNN(B)-frs(5240)-[135(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.128	70.78%		0.627	69.37%		0.613	72.19%	
42	DNN(B)-frs(5240)-[135(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.303	70.61%		0.612	71.03%		0.609	72.60%	
20	DNN(B)-frs(5240)-[90(relu)-85(relu)-109(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.154	70.61%		0.630	69.78%		0.611	71.85%	
12	DNN(B)-frs(5240)-[240(relu)-180(relu)-64(relu)-32(relu)-24(relu)-1(sigmoid)]-[ep(30)]	30	40	1.370	70.53%		0.631	69.70%		0.624	70.94%	
18	DNN(B)-frs(5240)-[90(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.023	70.53%		0.616	69.70%		0.612	72.19%	
64	DNN(B)-frs(5240)-[135(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.299	70.45%		0.622	69.21%		0.611	72.76%	
36	DNN(B)-frs(5240)-[120(relu)-85(relu)-109(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.226	70.36%		0.613	69.78%		0.613	71.11%	
6	DNN(B)-frs(5240)-[85(relu)-64(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	0.739	70.36%		0.608	71.27%		0.609	69.95%	
55	DNN(B)-frs(5240)-[120(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.139	70.36%		0.616	69.78%		0.612	72.19%	
9	DNN(B)-frs(5240)-[240(relu)-120(relu)-85(relu)-64(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.315	70.28%		0.627	69.04%		0.625	71.44%	
1	DNN(B)-frs(5240)-[20(relu)-12(relu)-8(relu)-1(sigmoid)]-[ep(30)]	30	40	0.569	71.44%		0.606	70.86%		0.616	70.12%	
11	DNN(B)-frs(5240)-[240(relu)-180(relu)-150(relu)-80(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.470	70.28%		0.636	69.21%		0.618	71.61%	
48	DNN(B)-frs(5240)-[120(relu)-85(relu)-109(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.193	70.28%		0.616	69.78%		0.610	71.94%	
0	DNN(B)-frs(5240)-[12(relu)-8(relu)-1(sigmoid)]-[ep(30)]	30	40	0.575	70.28%		0.611	70.12%		0.610	70.20%	
13	DNN(B)-frs(5240)-[280(relu)-240(relu)-180(relu)-64(relu)-32(relu)-24(relu)-1(sigmoid)]-[ep(30)]	30	40	1.550	70.28%		0.698	57.70%		0.689	57.70%	
32	DNN(B)-frs(5240)-[120(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.383	70.20%		0.618	69.54%		0.609	72.43%	
38	DNN(B)-frs(5240)-[135(relu)-85(relu)-64(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.254	70.20%		0.618	69.70%		0.615	71.44%	
27	DNN(B)-frs(5240)-[105(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.137	70.12%		0.613	70.86%		0.612	71.69%	
44	DNN(B)-frs(5240)-[135(relu)-85(relu)-109(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.366	70.03%		0.615	69.87%		0.613	72.19%	
41	DNN(B)-frs(5240)-[135(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.143	70.03%		0.607	71.52%		0.616	71.77%	
26	DNN(B)-frs(5240)-[105(relu)-85(relu)-94(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	20	1.184	70.03%		0.614	69.54%		0.608	71.19%	
21	DNN(B)-frs(5240)-[90(relu)-85(relu)-109(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.081	69.95%		0.612	71.03%		0.613	72.02%	
62	DNN(B)-frs(5240)-[135(relu)-85(relu)-79(relu)-32(relu)-1(sigmoid)]-[ep(30)]	30	40	1.323	69.95%		0.615	70.61%		0.610	71.85%	

Conclusion

- Satisfied performance: 73.8% test accuracy
- 29 features among Scenario Factors, Personal Attributes, Activity Frequency, Coupon-Related Factors
- help the business target customers who are more likely to accept the coupons, leading to enhanced customer engagement and sales.